Kinematics: velocity

Speed and Velocity

Going fast (**speed**) is not the same as knowing where you are going while moving fast (**velocity**).

- **Speed**: how fast something moves. Unit: $\frac{m}{s}$.
- **Velocity**: speed + the direction of movement. It is also measured in $\frac{m}{s}$, and the sign shows whether it moves to the right (+) or to the left (-).

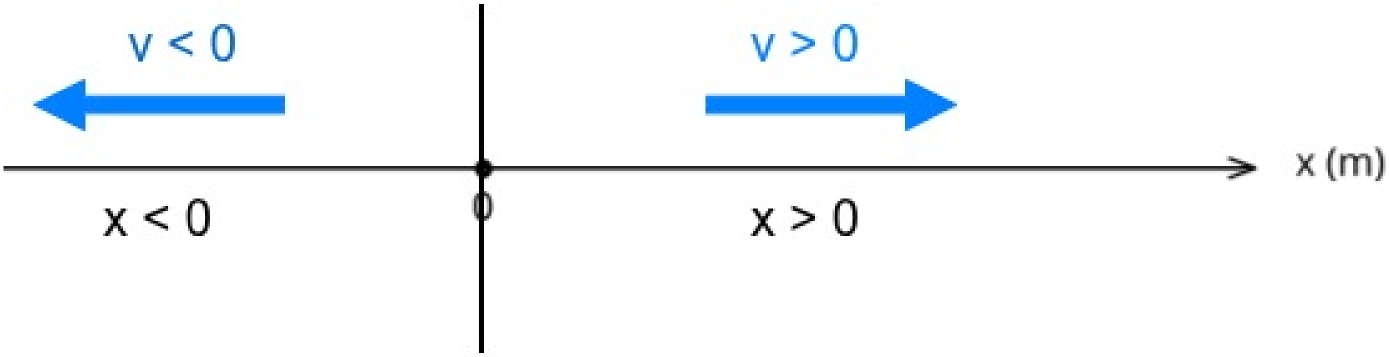

Figura 1: Sign rules for position and velocity

The *direction* tells us if the object moves to the right (positive velocity) or to the left (negative velocity) according to the sign rules shown above. It is shown with an arrow. The length of the arrow, called the *magnitude* (the numerical value without the sign), is the speed.

Normally, we will use the word **velocity**, but this means we must be careful with the signs:

- The displacement Δx can be positive or negative.
- **Time is always positive**.

Rearranging the Formula

❗ Very important

It is very important to know how to calculate velocity and how to rearrange the formula for all the variables.

- Velocity:
  $v = \frac{s}{t}$. It is measured in **metres per second ($\frac{m}{s}$)**.
- Rearranging to find distance:
  $v = \frac{s}{t} \rightarrow v \cdot t = s$. It is measured in **metres (m)**.
- Rearranging to find time:
  $v = \frac{s}{t} \rightarrow v \cdot t = s \rightarrow t = \frac{s}{v}$. It is measured in **seconds (s)**.

Units of Velocity: $\frac{m}{s}$ and $\frac{km}{h}$

In general, it is better to write $\dfrac{m}{s}$ instead of m/s and $\dfrac{km}{h}$ instead of km/h because it is clearer which part is the denominator.

It is very useful to learn how to change between different velocity units. We do this with conversion factors.

Converting from $\dfrac{m}{s}$ to $\dfrac{km}{h}$

$$1\frac{m}{s} = \frac{1\ m}{s} \cdot \frac{1\ km}{1000\ m} \cdot \frac{60 \cdot 60\ s}{1\ h} = \frac{3600\ km}{1000\ h} = 3.6\frac{km}{h}$$

Converting from $\dfrac{km}{h}$ to $\dfrac{m}{s}$

$$1\frac{km}{h} = \frac{1\ km}{h} \cdot \frac{1000\ m}{1\ km} \cdot \frac{1\ h}{3600\ s} = \frac{1000\ m}{3600\ s} = \frac{1}{3.6}\frac{m}{s}$$

Unit Conversion Exercises

1. Convert 15 m/s into km/h.
2. Convert 90 km/h into m/s.

Solutions

1. Convert $15\ m/s$ **into** km/h.

$$15\frac{m}{s} \cdot \frac{1\,km}{1000\,m} \cdot \frac{3600\,s}{1\,h}$$

$$= 15 \cdot \frac{3600}{1000}\frac{km}{h}$$

$$= 15 \cdot 3.6 = 54\frac{km}{h}$$

- *Answer:* $15\dfrac{m}{s} = 54\dfrac{km}{h}$.

2. Convert $90\ km/h$ **into** m/s.

$$90\frac{km}{h} \cdot \frac{1000\,m}{1\,km} \cdot \frac{1\,h}{3600\,s}$$

$$= 90 \cdot \frac{1000}{3600}\frac{m}{s}$$

$$= \frac{90}{3.6} = 25\frac{m}{s}$$

- *Answer:* $90\frac{km}{h} = 25\frac{m}{s}$

Instantaneous Velocity and Average Velocity

- **Instantaneous velocity**: the velocity an object has at one exact moment during its movement. You can see this in cars: the dashboard shows the car's velocity at each moment.
- **Average velocity**: the total distance travelled divided by the total time taken. In all our problems we will use this idea of velocity.

Example: the distance from Burgos to Palencia is almost 90 km. A car cannot travel at exactly the same velocity all the time because of traffic lights or different speed limits. But if we divide the total distance by the total time, we get the average velocity:

$$d = 90kmt = 1hv_m = v = \frac{d}{t} = \frac{90}{1} = 90\frac{km}{h}$$

Velocity Problems

1. The Castellana Bus.
An EMT bus travels the **6 km** between Plaza de Colón and the Cuatro Torres. If the journey is in a straight line (without stops) and takes exactly **15 minutes** (0.25 hours), what is the average velocity of the bus in km/h?

Solution

Using the formula:
$v = \frac{d}{t}$
$v = \frac{6 \text{ km}}{0.25 \text{ h}} = 24 \text{ km/h}$

2. The Scooter in Retiro Park.
A boy rides his electric scooter along the Paseo de Fernán Núñez in Retiro Park, which is **750 metres** long. If he takes **150 seconds** to complete the journey at a constant speed, what is his velocity in metres per second?

Solution

$v = \frac{d}{t}$
$v = \frac{750 \text{ m}}{150 \text{ s}} = 5 \text{ m/s}$

3. Madrid Underground (Line 1).
A metro train is at position $x_i = 200$ m (taking Puerta del Sol as the origin). A few seconds later, it is at position $x_f = 1400$ m (near Bilbao station). If it takes **80 seconds** to travel from one point to the other, first calculate the displacement and then the velocity in m/s.

Solution

1. Displacement (Δx):

$$x_f - x_i = 1400 - 200 = 1200 \text{ m}$$

2. Velocity:

$$v = \frac{\Delta x}{t} = \frac{1200 \text{ m}}{80 \text{ s}} = 15 \text{ m/s}$$

Additional Resources

- English video Motion in one dimension
- Straight-line motion